

# ECEN 4517/5517

## Power Electronics and Photovoltaic Power Systems Laboratory

Condensed Combined Lab Report — Experiments 2–5

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### System Overview

This report summarizes Experiments 2 through 5 of the ECEN 4517/5517 laboratory sequence, collectively implementing a complete battery-backed photovoltaic (PV) power system. The full signal chain is:

**PV Panel  $\rightarrow$  Buck DC-DC (MPPT/Charge)  $\rightarrow$  12 V Battery  $\rightarrow$  Cascaded Boost (150 V)  $\rightarrow$  Modified Sine-Wave Inverter (120 V<sub>rms</sub>)**

- **Exp. 2:** Characterization of the Shell SQ-85P PV panel and direct-energy-transfer (DET) battery charging.
- **Exp. 3 (Part 1):** Design and open-loop testing of a 100 kHz buck DC-DC converter.
- **Exp. 3 (Part 2):** Closed-loop MPPT (Perturb & Observe) and battery overcharge protection.
- **Exp. 4 (Part 1):** Design and open-loop testing of a cascaded boost converter (12 V  $\rightarrow$  150 V).
- **Exp. 4 (Part 2):** Integral feedback compensator for 150 V regulation.
- **Exp. 5:** Full H-bridge modified sine-wave inverter (150 V<sub>dc</sub>  $\rightarrow$  120 V<sub>rms</sub>, 60 Hz).

## 1 Experiment 2: PV Panel, Battery, and Direct Energy Transfer

### 1.1 PV Cart and Battery

The mobile PV cart (Figure ??) provides isolated DC supplies and a sine-wave inverter for test equipment. The battery is a 56 Ah, 12 V valve-regulated lead-acid (VR-AGM) unit; 100% SOC corresponds to  $\geq 12.80$  V open-circuit.

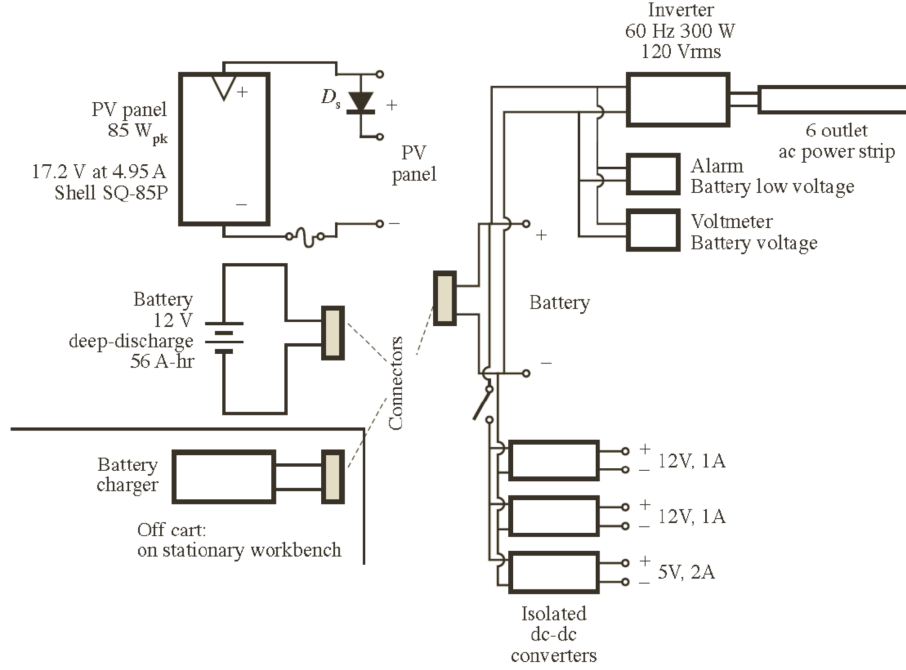


Figure 1: Electrical diagram of the laboratory mobile PV cart.

## 1.2 PV Panel Characterization

The Shell PowerMax Ultra 85-P panel (36 series silicon cells,  $P_{max} = 85 \text{ W}$ ,  $V_{mp} = 17.2 \text{ V}$ ,  $I_{mp} = 4.95 \text{ A}$  at STC) was characterized outdoors at  $1164 \text{ W/m}^2$ . Key results are shown in Table ??.

| Parameter            | STC (Nameplate) | Measured ( $1164 \text{ W/m}^2$ ) |
|----------------------|-----------------|-----------------------------------|
| $V_{oc}$ (V)         | 22.2            | 21.50                             |
| $I_{sc}$ (A)         | 5.45            | 5.22                              |
| $V_{mp}$ (V)         | 17.2            | 16.90                             |
| $I_{mp}$ (A)         | 4.95            | 4.75                              |
| $P_{mp}$ (W)         | 85              | 80.28                             |
| Panel Efficiency (%) | 13.44           | 10.91                             |

Table 1: PV panel nameplate vs. measured characteristics.

The measured efficiency (10.91%) is lower than the STC rating (13.44%) due to elevated cell temperature and panel aging. Shading just four of the 36 cells reduced peak power from 80.28 W to 33.93 W, a 58% loss — confirming the importance of bypass diodes. The measured I-V curve is shown in Figure ??.

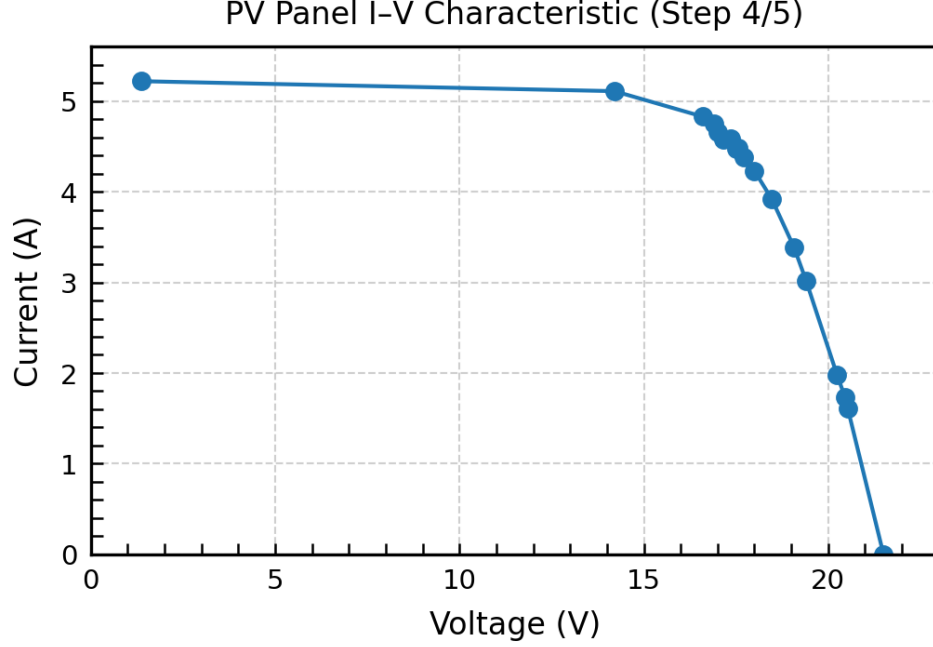


Figure 2: Measured  $I_{pv}$ - $V_{pv}$  characteristic, unshaded panel at  $1164 \text{ W/m}^2$ . MPP at  $V_{mp} = 16.90 \text{ V}$ ,  $P_{mp} = 80.28 \text{ W}$ .

The equivalent circuit model (photocurrent source  $I_0$ , diode  $I_{D0}$ ,  $R_s$ ,  $R_p$ ) was extracted via iterative MATLAB fitting:  $k_i = 5.24 \times 10^{-3}$ ,  $I_{D0} = 529 \text{ pA}$ ,  $R_p = 2.54 \Omega$ ,  $R_s = 11.24 \text{ m}\Omega$ . Model predictions agree with measurements to within 6%.

### 1.3 Direct Energy Transfer (DET) Charging

With the PV panel connected directly to the battery through a series diode, the panel was forced to operate at the battery voltage ( $14.13 \text{ V}$ ), delivering  $71.08 \text{ W}$  — approximately 11.5% below the  $80.28 \text{ W}$  MPP. This motivates the MPPT converter in Experiment 3.

## 2 Experiment 3 Part 1: Open-Loop Buck DC-DC Converter

### 2.1 System and Design

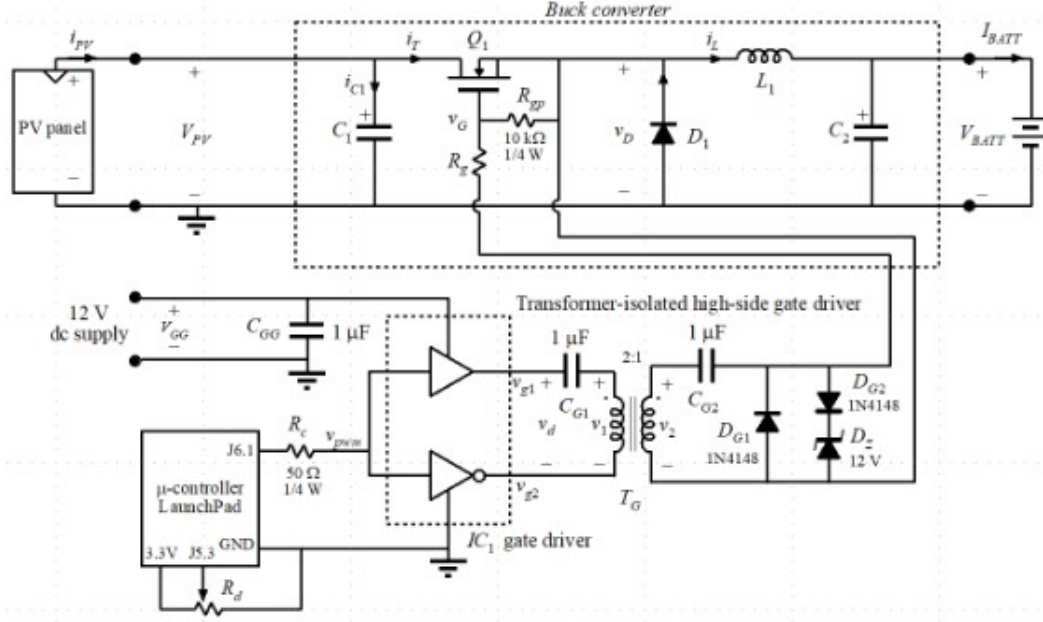


Figure 3: Buck DC-DC converter system block diagram (PV panel  $\rightarrow$  buck  $\rightarrow$  battery).

A 100 kHz buck converter was designed and constructed on protoboard. Key components: HUF75321P3 MOSFET, MBR1645 Schottky diode, TC4428A gate driver, custom-wound ferrite inductor ( $L = 22.23 \mu\text{H}$ , N95, 23 turns, #18 AWG, 3 mm air gap, target  $22 \mu\text{H}$ , 1.05% error).

### 2.2 Performance Results

| Test Condition                   | $V_{in}$ (V) | $V_{out}$ (V) | $\eta$ (%) |
|----------------------------------|--------------|---------------|------------|
| Full load ( $\approx 85$ W, CCM) | 22.22        | 11.12         | 91.3       |
| Medium load (15 W, CCM)          | 22.22        | 13.07         | 96.7       |
| Light load (1 W, DCM)            | 25.13        | 13.10         | 83.0       |

Table 2: Buck converter performance at three load levels. Transition from CCM to DCM is evident at 1 W.

Outdoor open-loop sweep confirmed that maximum battery charging power (53.92 W at  $I_{batt} = 4.42$  A) occurred near  $D = 75\%$ , corresponding to  $V_{pv} = 17.64$  V — close to the theoretical MPP. Measured ripple (3.40 A) slightly exceeded the design target (2.70 A), attributed to partial core saturation at high current.

## 3 Experiment 3 Part 2: MPPT and Charge Control

### 3.1 Sensing and Control Architecture

Battery current is sensed via a  $20 \text{ m}\Omega$  shunt amplified by an INA290A2 (gain = 50 V/V), followed by a resistor divider for a net gain of 22.93 V/V. Battery voltage is sensed with a  $10 \text{ k}\Omega/3.3 \text{ k}\Omega$  divider

( $K_V = 0.25$ ). Both signals are digitized by the microcontroller ADC (3.3 V, 12-bit). The full schematic with sensing circuitry is shown in Figure ??.

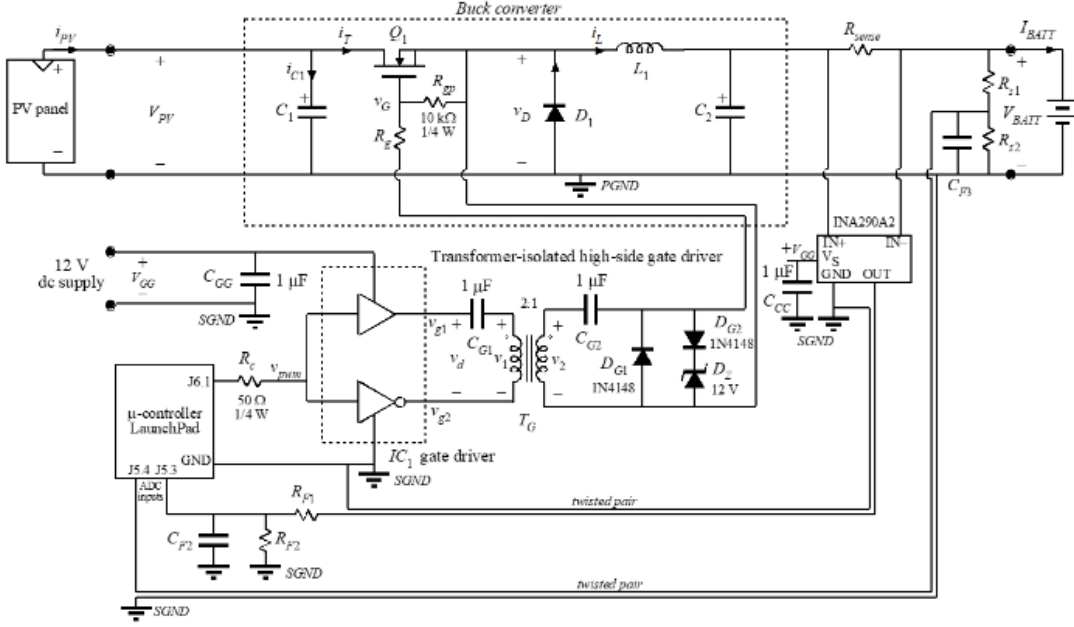


Figure 4: Buck DC-DC converter with battery current and voltage sensing.

### 3.2 Perturb & Observe MPPT Algorithm

The P&O algorithm perturbs duty cycle by  $\Delta D = 4$  every 100 ms, comparing consecutive PV power measurements. Step size is divided by cumulative battery current to improve convergence at low-irradiance conditions. Charge control disables the converter at  $V_{batt} \geq 12.8$  V and resumes at  $V_{batt} \leq 12.6$  V.

### 3.3 Outdoor MPPT Results

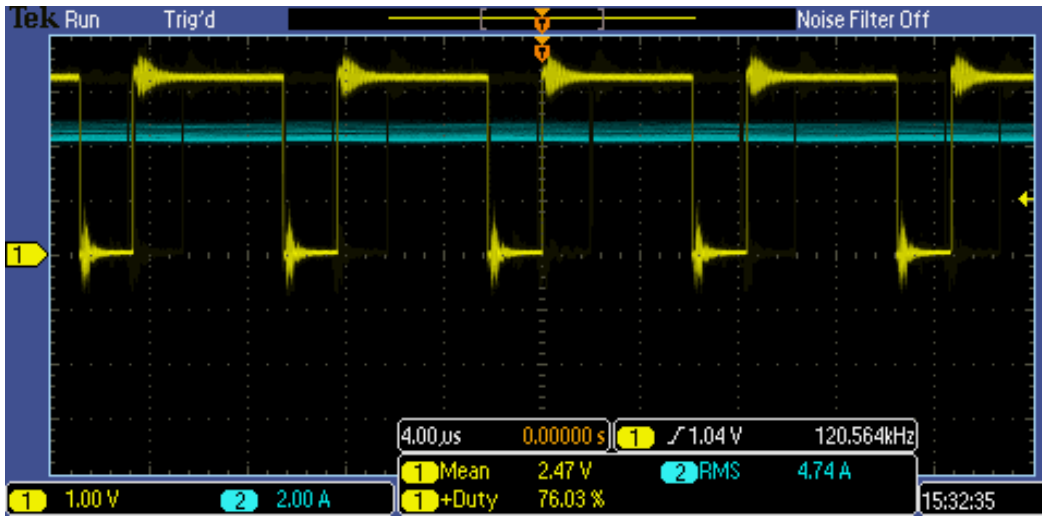


Figure 5: PV voltage and filtered PWM duty cycle during P&O MPPT outdoors ( $1026 \text{ W/m}^2$ , unshaded).

| Parameter            | Direct Transfer (Exp. 2) | MPPT (Exp. 3P2) |
|----------------------|--------------------------|-----------------|
| $V_{pv}$ (V)         | 14.13                    | 18.10           |
| $P_{pv}$ (W)         | 71.08                    | 51.40           |
| $P_{batt}$ (W)       | 63.48                    | 50.25           |
| Converter $\eta$ (%) | —                        | 97.76           |

Table 3: MPPT vs. direct energy transfer comparison (unshaded). Lower MPPT power attributed to atypically low panel current on the test day.

Shading four cells while using MPPT reduced output to 1.47 W (97.1% power loss), highlighting the challenge of partial shading with single-peak P&O trackers.

## 4 Experiment 4 Part 1: Cascaded Boost Converter (Open Loop)

### 4.1 Design Overview

Two cascaded boost stages step the 12 V battery bus to 150 V<sub>dc</sub>:

$$M = \frac{V_{OUT}}{V_{BATT}} = \frac{1}{(1 - D)^2}, \quad D = 71.18\%$$

An SG3525A analog PWM IC generates complementary gate signals at  $f_s = 94.34$  kHz (simulated: 108.61 kHz, 15% error due to component tolerance). Gate drivers are TC4428A ICs. Key power components:  $Q_1$  PSMN013-100PS (100 V),  $Q_2$  FQP11N40C (400 V),  $D_1$  MBR1645 (45 V Schottky),  $D_2$  FES8GT (400 V ultrafast). Custom inductors:  $L_1 = 42$   $\mu$ H (target 47  $\mu$ H, 11.9% error) and  $L_2 = 365$   $\mu$ H (target 550  $\mu$ H, 50.7% error — largest source of excess ripple).

### 4.2 Measured Performance

| Parameter               | Value                          |
|-------------------------|--------------------------------|
| Input voltage $V_g$ (V) | 12.08                          |
| Output voltage (V)      | 151.1                          |
| Duty cycle $D$ (%)      | 71.18                          |
| Efficiency $\eta$ (%)   | 79.0                           |
| $Q_1$ peak $V_{DS}$ (V) | 62 (rated 100 V, 61% margin)   |
| $Q_2$ peak $V_{DS}$ (V) | 174 (rated 400 V, 130% margin) |

Table 4: Open-loop cascaded boost converter operating point and stress summary.

Both MOSFETs remain well within voltage ratings. The 79% efficiency is lower than the buck stage primarily due to the larger  $R_{DS(on)}$  of  $Q_2$  and the under-wound  $L_2$ .

## 5 Experiment 4 Part 2: Closed-Loop Feedback Controller

### 5.1 Control Design

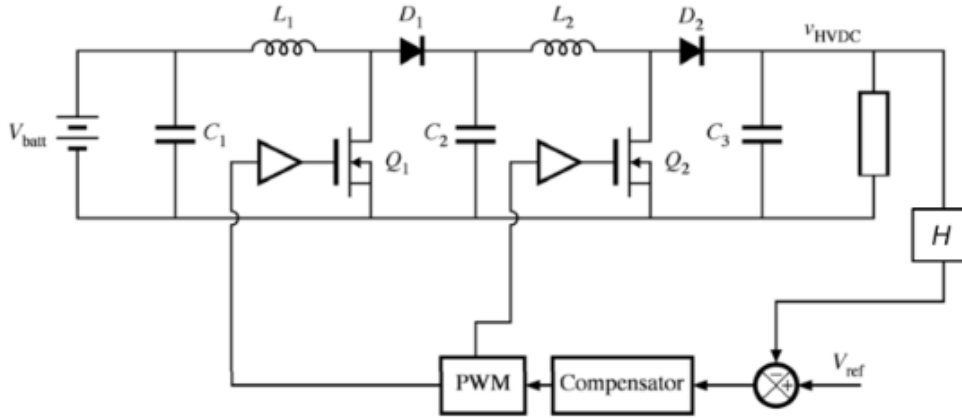


Figure 6: Closed-loop cascaded boost converter block diagram with integral compensator.

The control-to-output transfer function  $G_{vc}(s)$  was measured with a Bode 100 network analyzer at the nominal 150 V operating point. The measured DC gain is 55 dB with a dominant resonant pole at 470 Hz and a RHP zero near 10 kHz. An integral (Type 1) compensator was designed:

$$G_c(s) = -\frac{1}{s R_2 C_c}, \quad R_2 = 10 \text{ k}\Omega, \quad C_c = 1 \mu\text{F}$$

Output voltage setpoint:

$$V_{HVDC} = V_{ref} \cdot \frac{R_1 + R_2}{R_2} = 2.55 \times \frac{570 \text{ k}\Omega}{10 \text{ k}\Omega} \approx 145.4 \text{ V}$$

## 5.2 Measured Stability and Regulation

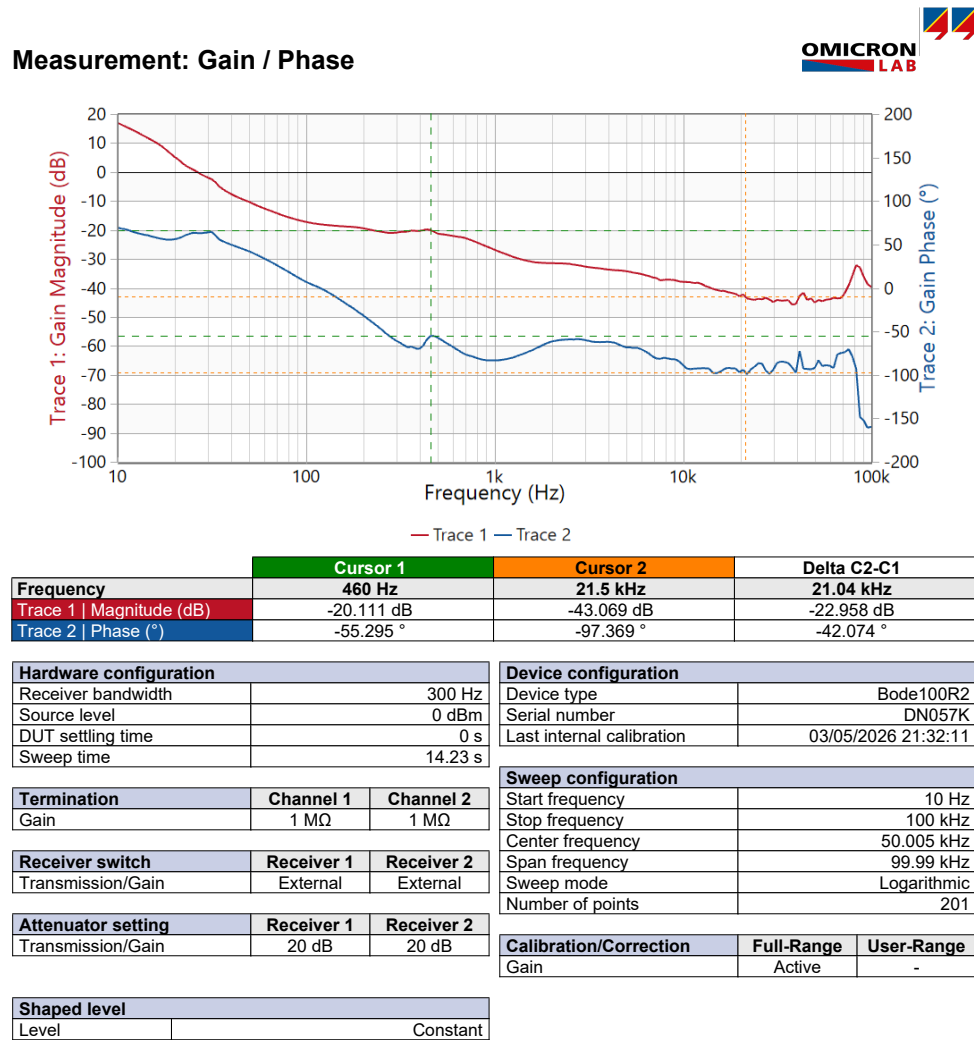


Figure 7: Measured loop-gain Bode plot (Bode 100). Crossover at 30 Hz, phase margin = 245°.



| Parameter                                          | Simulated | Measured |
|----------------------------------------------------|-----------|----------|
| Crossover $f_c$ (Hz)                               | 390       | 30       |
| Phase margin PM ( $^\circ$ )                       | 111       | 245      |
| Line regulation ( $V_g = 11\text{--}14\text{ V}$ ) | —         | 0.1%     |
| Start-up time (s)                                  | 1.0       | 0.95     |

Table 5: Closed-loop stability and regulation performance.

The low crossover frequency (30 Hz) provides generous phase margin ( $245^\circ$ ) and excellent line regulation (0.1% over 11–14 V input range), consistent with the integrator’s infinite DC gain.

## 6 Experiment 5: Modified Sine-Wave DC-AC Inverter

### 6.1 System and Design

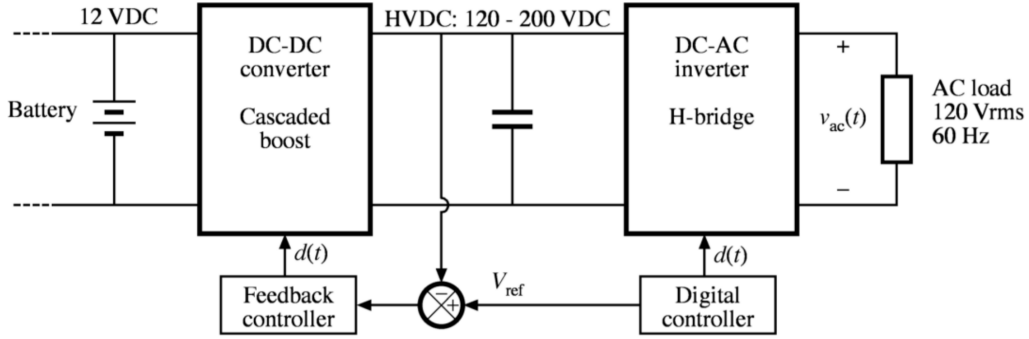


Figure 8: Complete inverter system block diagram (Experiments 4 and 5 combined).

The inverter is a full H-bridge (four IRF730 MOSFETs, 400 V/5.5 A) driven by two IR21094 half-bridge gate-driver ICs. A  $C_{dc} = 33\text{ }\mu\text{F}$ , 250 V capacitor decouples switching transients from the boost converter bus. The Arduino UNO generates complementary 60 Hz gate signals with duty cycle:

$$D = \left( \frac{V_{ac,RMS}}{V_{HVDC}} \right)^2 = \left( \frac{120}{150} \right)^2 = 0.64$$

so that  $V_{ac,RMS} = \sqrt{D} \cdot V_{HVDC} = \sqrt{0.64} \times 150 = 120\text{ V}_{rms}$ .

## 6.2 Measured Performance



Figure 9: Oscilloscope capture of modified sine-wave output  $v_{ac}(t)$  at 60 Hz.

| Parameter             | Expected | Measured |
|-----------------------|----------|----------|
| $V_{HVDC}$ (V)        | 150      | 149.10   |
| Duty cycle $D$        | 0.64     | 0.68     |
| Output frequency (Hz) | 60       | 59.9     |
| $V_{ac,RMS}$ (V)      | 120      | 119      |
| Output power (W)      | 60       | 42       |

Table 6: Expected vs. measured inverter performance (resistive lightbulb load).

The measured duty cycle of 0.68 (vs. 0.64 design) reflects the slight increase needed to overcome conduction losses in the MOSFETs and gate drivers. Output voltage ( $119 V_{rms}$ ) and frequency (59.9 Hz) are within 1% of target.

## 7 Overall System Conclusions

| Stage             | Function                                 | Key Result                                        | $\eta$ |
|-------------------|------------------------------------------|---------------------------------------------------|--------|
| PV Panel (SQ-85P) | Energy source                            | $P_{mp} = 80.3 \text{ W}$ at $1164 \text{ W/m}^2$ | 10.9%  |
| Buck + P&O MPPT   | MPPT & charge ctrl                       | $D = 71.2\%$ , $P_{batt} = 50.3 \text{ W}$        | 97.8%  |
| Cascaded Boost    | $12 \text{ V} \rightarrow 150 \text{ V}$ | $V_{out} = 149 \text{ V}$ , $\Delta V < 0.1\%$    | 79–92% |
| H-Bridge Inverter | DC-to-AC inversion                       | $119 V_{rms}$ at 59.9 Hz                          | —      |

Table 7: Summary of all four laboratory stages.

The system demonstrates a functional end-to-end PV power chain from solar irradiance to  $120 V_{rms}$  AC output. Key findings:

- Partial shading (4/36 cells) reduced PV output by 58% in DET mode, motivating MPPT.

- The P&O MPPT algorithm achieved 97.76% converter efficiency under clear-sky conditions; performance degraded severely under shading due to multiple I-V curve peaks.
- Integral feedback compensation provided 0.1% line regulation across the 11–14 V battery input range, with  $245^\circ$  phase margin.
- The modified sine-wave inverter produced  $119 V_{\text{rms}}$  at 59.9 Hz, within 1% of specifications, completing the battery-to-AC signal path.